

Mackenzie Sampson

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Education

University of British Columbia – Bachelor of Computer Science

Thompson Rivers University – Bachelor of Business Administration, Finance

Skills

Languages: Go, C++, Python, SQL, C#, TypeScript

Databases: PostgreSQL, Prisma, DynamoDB

Cloud & Infrastructure: AWS (Lambda, API Gateway, CDK, DynamoDB, SQS, EventBridge), Docker, Linux

Experience

Software Engineer, PocketCaps – Vancouver, BC Mar 2021 – 2025

- Founded and ran a keycap e-commerce platform processing multiple six figures across 5,000+ customers.
- Prevented silent oversell by guarding inventory decrements against negative stock, alerting operators on failure.
- Designed a 19-model PostgreSQL schema exposing 40+ REST endpoints to a decoupled storefront client.
- Eliminated permanent order loss on Stripe webhook retries.

Technical Artist, Tools & Pipeline (Contract), Electronic Arts – Burnaby, BC Sept 2019 – Nov 2020

- Owned the Ignite pipeline porting FIFA 21 to Switch, integrating lower LODs and reduced textures from HD.
- Built a Maya tool that reimported hair bones from HD upstream and reweighted them for the Switch build.
- Cut artist processing time 80% with Python/Maya tooling correcting map channel assignments and naming.
- Built tooling against the CMS to drive per-market localizations stripping gambling and alcohol sponsors.

Software Compliance Intern, SAP – Vancouver, BC May – Dec 2017

- Protected SAP from license violations by scanning products with Black Duck alongside legal and dev teams.
- Cut data entry time and errors ~85% with a Python/Selenium tool that ingested spreadsheets into Black Duck.
- Shortened audit reporting cycles by standardizing compliance workflows across departments.

Software Engineer Intern, MDA – Richmond, BC Jan – Sept 2016

- Delivered a CMS migration of MDA's corporate web platform, validating page parity before cutover.
- Shipped front-end features on a small team, turning stakeholder requirements into deployed changes.

Projects

Inventory Reservation Service | Go, PostgreSQL github.com/macsampson/reservation-service

- Standalone Go service that reserves stock at cart time, closing the oversell window before payment.
- Shares a Postgres instance with Stockroom under column-ownership invariants, so both write concurrently.

Stockroom | TypeScript, Next.js, PostgreSQL github.com/macsampson/stockroom

- Generalizing the PocketCaps platform into a self-hostable commerce CMS for arbitrary stores.
- Migrated storage and database layers (Supabase/Cloudinary → Neon/Vercel Blob) with CI and one-click deploy.

Ghosted | TypeScript, React, AWS CDK, Lambda, DynamoDB github.com/macsampson/ghosted

- Built a live app tracking job-application status by classifying Gmail messages with an LLM.
- Provisioned the full AWS backend as code with CDK v2 on a single-table DynamoDB design.

Monte Carlo Path Tracer | C++, Intel TBB, Callgrind github.com/macsampson/pathtracer

- Physically-based renderer; replaced linear ray-scene intersection with a BVH for a $4.7\times$ speedup.
- Parallelized with Intel TBB for a $27\times$ end-to-end speedup, 3m 28s → 7.6s on a fixed benchmark.
- Profiled with Callgrind, then fixed degenerate AABBs, cutting intersection tests from 42% to 16% of runtime.